

Module 23.09: What is the Cost Function? (MSE & the $1/(2m)$ Factor)

1. Loss Function vs. Cost Function

In machine learning terminology, it is vital to distinguish between a **Loss Function** and a **Cost Function**:

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| Single Training Example  $(x^{(i)}, y^{(i)})$  ---> Loss Function:  $L(y, \hat{y})$  |
| Entire Dataset of  $m$  Examples ---> Cost Function:  $J(w, b)$  |
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- **Loss Function** $L(\hat{y}^{(i)}, y^{(i)})$: Quantifies the discrepancy for a **single** individual training instance i . For squared loss:

$$L(\hat{y}^{(i)}, y^{(i)}) = \frac{1}{2} (\hat{y}^{(i)} - y^{(i)})^2$$

- **Cost Function** $J(w, b)$: Measures the aggregate performance of the model parameters across the **entire dataset** of m training instances. It is the empirical risk over the sample distribution.
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2. Mathematical Definition of the Linear Regression Cost Function

For standard univariate Linear Regression with hypothesis $\hat{y}^{(i)} = wx^{(i)} + b$, the cost function is defined as:

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)})^2 = \frac{1}{2m} \sum_{i=1}^m (wx^{(i)} + b - y^{(i)})^2$$

3. Mathematical Rationale for the $\frac{1}{2m}$ Factor

Why do we divide by $2m$ instead of merely minimizing the raw sum of squared errors $\sum (y - \hat{y})^2$?

1. The $\frac{1}{m}$ Factor (Sample Size Invariance)

- If we do not divide by m , the total error increases linearly as more data points are added ($m \rightarrow 10^6 \implies \text{Error} \rightarrow \infty$).
- Dividing by m computes the **average** squared error per observation.
- This makes the cost magnitude and learning rate α robust and invariant to the size of the training set.

2. The $\frac{1}{2}$ Factor (Mathematical Elegance in Calculus)

- When minimizing $J(w, b)$ via gradient descent or analytical differentiation, we must take the derivative with respect to parameters:

$$\frac{\partial}{\partial w} \left[\frac{1}{2} (u)^2 \right] = \frac{1}{2} \cdot 2u \cdot \frac{\partial u}{\partial w} = u \cdot \frac{\partial u}{\partial w}$$

- The constant coefficient $\frac{1}{2}$ cleanly cancels out the exponent 2 that drops down via the power rule of calculus!
- Without $\frac{1}{2}$, every gradient update formula would carry an extraneous factor of 2, complicating calculations without altering the location of the minimum:

$$\arg \min J(w, b) = \arg \min \left(\frac{1}{2m} \sum_{i=1}^m (e^{(i)})^2 \right) = \arg \min \left(\sum_{i=1}^m (e^{(i)})^2 \right)$$

4. Cost as a Function of Parameters

It is essential to understand what is fixed and what varies in $J(w, b)$:

- **Fixed Quantities:** The empirical dataset $(x^{(1)}, y^{(1)}), \dots, (x^{(m)}, y^{(m)})$ consists of fixed historical constants.
- **Independent Variables:** The weights w and b are the variables. The cost function maps parameter space $\mathbb{R}^2$ to a non-negative real scalar $\mathbb{R}_{\geq 0}$:

$$J: \mathbb{R}^2 \rightarrow \mathbb{R}_{\geq 0}$$

The optimization objective of linear regression is formalized as:

$$(w^*, b^*) = \arg \min_{w, b} J(w, b)$$